

Vivian Ha

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Profile

AI/ML master by research student with hands-on experience in deep learning, neural networks, and intelligent systems development. Proven expertise in TensorFlow, MATLAB, and end-to-end machine learning with a strong foundation in software design, development and implementation. Quick to master emerging AI technologies and development frameworks with demonstrated ability in research implementation and software deployment.

Education

Master of Philosophy (Engineering) | The University of Sydney | 2025 – present

- Thesis: “End-to-End MU-MIMO with Neuromorphic Receiver for Energy-Efficient IoT Applications”

Bachelor of Engineering First Class Honors (Software) | The University of Sydney | 2021 – 2025

Professional Experience

1. R&D Intern | Audinate | Dec 2023 – Feb 2024

- Contributed to commercial IoT cloud solution architecture and prototyping
- Designed and developed GraphQL API workflows for device management in Dante Cloud

2. Engineering Research Scholarship | The University of Sydney | Dec 2023 – Feb 2024; Dec 2024 – Feb 2025

- Optimized AI-assisted wireless communication systems using end-to-end supervised learning and neural network
- Implemented autoencoder-based solutions and generative models to improve transmission in wireless systems

Key Projects [\[GitHub\]](#)

1. 5G End-to-End Wireless Communication System with CWGAN-based Channel | Thesis Research *TensorFlow, NVIDIA Sionna, Python*

- Developed end-to-end learning system with neural mapper and deep CNN receiver
- Achieve equivalent transmission efficiency as a known wireless channel with a GAN-based channel surrogate.

2. Atlassian Rovo GenAI Jira Assistant | Atlassian ForgeHACK *JavaScript, Atlassian Forge, Atlassian Rovo AI, REST*

- Developed a Rovo Agent application that achieved sub-3s response for intelligent meeting notes preparation through automated prompt engineering

3. AWS Serverless Image App *AWS Lambda, S3, EC2, RDS, Python Flask*

- Designed event-driven serverless architecture with S3-triggered automated image analysis and caption generation, achieving 75% success rate under high load (500 req/sec) testing
- Implemented three-tier security model with auto-scaling EC2 instances and multi-AZ deployment across VPC subnets, integrating external AI APIs through NAT gateway.

Skills

- **AI/ML & Data analytics:** TensorFlow, MATLAB, Deep Learning & Neural Networks
- **Cloud & AI Services:** AWS Lambda, S3, EC2, Serverless Architecture
- **Programming:** Python, Java, C, JavaScript/TypeScript, MATLAB
- **Development & Databases:** React, GraphQL, Spring Boot, REST APIs, Flask, PostgreSQL, MySQL
- **Tools & Platforms:** Git, Jenkins, Jira, Bitbucket, Android Studio